The empirical recurrence for OEIS A220726, proved by transfer matrix

Adrian Perez Fontelles
Independent researcher

2 September 2026

Abstract

OEIS A220726 counts the ways of linking the cells of a grid 2 cells wide, each cell either to itself or to exactly 2 of its neighbours, the links being reciprocal, and carries an empirical recurrence of order 4 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. A reciprocal link is an undirected edge and a self-link is no edge, so what is counted is the spanning subgraphs of a grid graph in which every vertex has degree 0 or 2; the edges crossing the boundary between two consecutive rows are a state, the count is a walk count on $S = 8$ of them, and the conjectured recurrence is then settled by a finite exact computation.

2020 Mathematics Subject Classification. 05A15, 05C30, 05C70.

1 The conjecture

OEIS A220726 is "Number of ways to reciprocally link elements of a 2 X n array either to themselves or to exactly two horizontal, diagonal or antidiagonal neighbors.". It has offset 1 and begins

$$1, 2, 7, 21, 68, 221, 717, 2330, \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 3*a(n-1) + a(n-2) - 2*a(n-4).

As of the "Last modified" line on the live entry (Aug 02 2018 15:20:00, revision 8) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

A link is *reciprocal*: if u is linked to v then v is linked to u. So the links of a configuration are the edges of an undirected graph on the cells, and a cell "linked to itself" is a cell lying in no edge. The neighbour relation the entry names is the offset set

$$(-1, -1), (-1, 0), (-1, 1), (1, -1), (1, 0), (1, 1),$$

and the condition "either to themselves or to exactly 2 neighbours" says that in that graph every cell has degree 0 or 2. Hence $a(n)$ counts the spanning subgraphs of the grid graph on the offset set above, of n rows and 2 columns, whose every degree is 0 or 2: a disjoint union of cycles, since every vertex of the link graph has degree 0 or 2. The entry writes the array with n as the number of COLUMNS; transposing it exchanges the two coordinates of every neighbour offset, which is done once, before anything else, so that the walk always runs down the rows.

One point of the English has to be fixed rather than guessed: whether a cell may spend both of its links on the same neighbour. It may not. For the king-move neighbourhood the 1×2 and 2×2 counts are the degree-0-or-2 subgraphs of K_2 and of K_4, which are 1 and $1 + 4 + 3 = 8$ with simple edges and larger if a doubled edge were allowed; the entries of that family publish 1 and 8. The reading is in any case pinned entry by entry, because the model below is required to reproduce every term the entry publishes.

3 The count is a walk count

Every offset above changes the row index by at most one, so an edge either lies inside a row or crosses the boundary between two consecutive rows. Take as state the set of edges crossing one boundary. Placing a row means choosing the horizontal edges inside it and the edges it sends down to the next row; at that moment each of its cells knows all of its links — those arriving from the row above, those inside the row, those going down — so its degree, and any condition on the pair of links at a cell, is settled there and never revisited.

Reading a configuration from the top therefore gives a walk that begins at the empty boundary, takes one step per row, and ends at the empty boundary, and every such walk comes from exactly one configuration. With M the adjacency matrix of the digraph on the $S = 8$ reachable states and ι, τ the indicator vectors of the empty boundary,

$$a(n) \;=\; \iota^\top M^n \tau,$$

so a is C-finite. States are generated from the empty boundary outward, so only the reachable ones are ever built; the alignment of the walk index with the entry's own n was checked against its published terms rather than assumed.

4 The proof

Lemma 1. *Let $q(t) = t^4 - \sum_i c_i t^{4-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+4} - \sum_i c_i M^{j+4-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 2.

$$a(n) \;=\; 3\,a(n-1) + a(n-2) - 2\,a(n-4)$$

for every $n > 3$.

Proof. The vector $w = q(M)\tau$ was formed by 4 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 3 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. □

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 26 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A220726.

[2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[3] L. Lovász and M. D. Plummer, *Matching Theory*, AMS Chelsea, 2009.

[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.